

Chapter 10 supplementary material

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This document accompanies Chapter 9 of

Flora, D.B. (2026). *Statistical modelling methods: A guide for the behavioural and social sciences* (2nd ed.). Sage Publications.

Model-implied covariance matrix for path analysis models

To demonstrate the idea of model-implied covariances, consider the simple, linear regression model, here presented using path analysis notation:

$$Y_i = \gamma X_i + \zeta_i.$$

If this linear regression model is correct for a particular population, then we can take the variance of both sides of the equation:

$$\text{VAR}(Y) = \text{VAR}(\gamma X + \zeta),$$

then separate the right-hand side into two variance terms:

$$(S10.1) \text{VAR}(Y) = \text{VAR}(\gamma X) + \text{VAR}(\zeta).$$

Next, because of the variance of a random variable (e.g., X) multiplied by a constant (e.g., γ) equals the constant squared times the variance of the variable, we can re-express the equation as

$$\text{VAR}(Y) = \gamma^2 \text{VAR}(X) + \text{VAR}(\zeta).$$

Now, if the best-fitting linear regression slope is calculated as

$$\gamma = \text{COV}(Y, X) / \text{VAR}(X),$$

then

$$(S10.2) \text{COV}(Y, X) = \gamma \text{VAR}(X).$$

This simple-regression model has only two observed variables, Y and X , and so the population covariance matrix Σ for the variables in this model consists of only the variance of Y , the variance of X , and the covariance between them:

$$\begin{pmatrix} \text{VAR}(Y) & \text{COV}(Y, X) \\ \text{COV}(Y, X) & \text{VAR}(X) \end{pmatrix}.$$

Then, given Equations S10.1 and S10.2, the observed covariance matrix can be expressed in terms of the model parameters with

$$\begin{pmatrix} \text{VAR}(Y) & \text{COV}(Y, X) \\ \text{COV}(Y, X) & \text{VAR}(X) \end{pmatrix} = \begin{pmatrix} \gamma^2 \text{VAR}(X) + \text{VAR}(\zeta) & \gamma \text{VAR}(X) \\ \gamma \text{VAR}(X) & \text{VAR}(X) \end{pmatrix} = \begin{pmatrix} \gamma^2 \phi + \psi & \gamma \phi \\ \gamma \phi & \phi \end{pmatrix},$$

where $\phi = \text{VAR}(X)$ and $\psi = \text{VAR}(\zeta)$. This equation shows the expression of the population covariance matrix as a function of the model's free parameters, which are the regression slope (γ), the variance of the exogenous variable ($\phi = \text{VAR}(X)$), and the error variance ($\psi = \text{VAR}(\zeta)$).

Returning to the general matrix form of the path analysis model (i.e., Equation 10.1 in the main text), the model-implied covariance matrix Σ_θ can be broken down into (1) the implied covariances among the endogenous variables, $\Sigma_{YY|\theta}$, (2) the implied covariances among the exogenous variables, $\Sigma_{XX|\theta}$, and (3) the implied covariance of Y with X , $\Sigma_{YX|\theta}$. Specifically,

$$\Sigma_\theta = \begin{bmatrix} \Sigma_{YY|\theta} & \Sigma_{YX|\theta} \\ \Sigma_{YX|\theta} & \Sigma_{XX|\theta} \end{bmatrix},$$

where

$$\Sigma_{YY|\theta} = (\mathbf{I} - \mathbf{B})^{-1}(\mathbf{\Gamma}\mathbf{\Phi}\mathbf{\Gamma}' + \mathbf{\Psi})(\mathbf{I} - \mathbf{B})^{-1}',$$

$$\Sigma_{XX|\theta} = \mathbf{\Phi},$$

and

$$\Sigma_{YX|\theta} = \mathbf{\Phi}\mathbf{\Gamma}'[(\mathbf{I} - \mathbf{B})^{-1}]',$$

where $\mathbf{I}$ is a conformable identity matrix (see the online supplemental chapter on matrix algebra). Recall from Chapter 10 that $\mathbf{B}$ is a matrix of coefficients for the regression of endogenous variables on other endogenous variables, $\mathbf{\Gamma}$ is a matrix of coefficients for the regression of endogenous variables on exogenous variables, $\mathbf{\Psi}$ is the covariance matrix for the error terms in the scalar equations comprising the model, and $\mathbf{\Phi}$ is the covariance matrix for the observed exogenous variables. For details about the derivation of the model-implied covariance matrix, see Bollen (1989). Once again, the goal of estimation is to find numerical values for the free parameters so as to match these model-implied covariances to the observed sample covariances as closely as possible.

For the simple regression model presented above, $\Sigma_{YY|\theta}$ reduces to $\gamma^2\phi + \psi$, $\Sigma_{YX|\theta}$ reduces to $\gamma\phi$, and $\Sigma_{XX|\theta}$ is simply ϕ .

Model-implied covariance matrix for structural regression models

Similar to path analysis, the population model-implied covariance matrix Σ_{θ} for the full structural regression model can be broken down into (1) $\Sigma_{YY|\theta}$, which is the implied covariances among $\mathbf{Y}$, the observed indicators of the endogenous latent variables, (2) $\Sigma_{XX|\theta}$, which is the implied covariances among $\mathbf{X}$, the observed indicators of the exogenous latent variables, and (3) $\Sigma_{YX|\theta}$, which is the implied covariance of $\mathbf{Y}$ with $\mathbf{X}$. Specifically, these model-implied covariance matrices are:

$$\Sigma_{YY|\theta} = \mathbf{\Lambda}_Y(\mathbf{I} - \mathbf{B})^{-1}(\mathbf{\Gamma}\mathbf{\Phi}\mathbf{\Gamma}' + \mathbf{\Psi})(\mathbf{I} - \mathbf{B})^{-1}'\mathbf{\Lambda}_Y' + \mathbf{\Theta}_{\epsilon},$$

$$\Sigma_{XX|\theta} = \mathbf{\Lambda}_X\mathbf{\Phi}\mathbf{\Lambda}_X' + \mathbf{\Theta}_{\delta}, \text{ and}$$

$$\Sigma_{YX|\theta} = \mathbf{\Lambda}_Y(\mathbf{I} - \mathbf{B})^{-1}\mathbf{\Gamma}\mathbf{\Phi}\mathbf{\Lambda}_X',$$

here $\mathbf{\Lambda}_Y$ is the factor loading matrix for the measurement model of the observed endogenous variables (i.e., $\mathbf{Y}$), $\mathbf{B}$ is the matrix of coefficients for the regression of endogenous latent variables on other endogenous latent variables, $\mathbf{\Gamma}$ is the matrix of coefficients for the regression of endogenous latent variables on exogenous latent variables, $\mathbf{\Phi}$ is the covariance matrix for the exogenous latent variables, $\mathbf{\Psi}$ is the covariance matrix for the error terms in the structural regression equations, $\mathbf{\Theta}_{\epsilon}$ is the error covariance matrix for $\mathbf{Y}$, $\mathbf{\Lambda}_X$ is the factor loading matrix for the measurement model of the observed exogenous variables (i.e., $\mathbf{X}$), and $\mathbf{\Theta}_{\delta}$ is the error covariance matrix for $\mathbf{X}$. Again, see Bollen (1989) for further description.

Two-step rule, step 1:

Prosocial behaviour structural regression model recast as CFA model

The first step of the two-step rule of model identification for structural regression models is to rewrite the full model as a measurement model without any directional, structural regression effects. Specifically, the complete model is recast as a CFA model: All latent variables in both ξ and η are considered freely correlated with each other and there are no directional relationships among them (i.e., ignore the $\mathbf{\Gamma}$ and $\mathbf{B}$ matrices). Then, determine whether this CFA model would

be identified (there is no actual need to estimate it, though). Figure S10.1 shows the corresponding CFA model for the prosocial behaviour example. This CFA model does not satisfy either the two-indicator or three-indicator rules for CFA presented in Chapter 9. However, this CFA model is identified according to more complex criteria; recall that although the two- and three-indicator rules are sufficient, they are not necessary for establishing the identification of a CFA model.

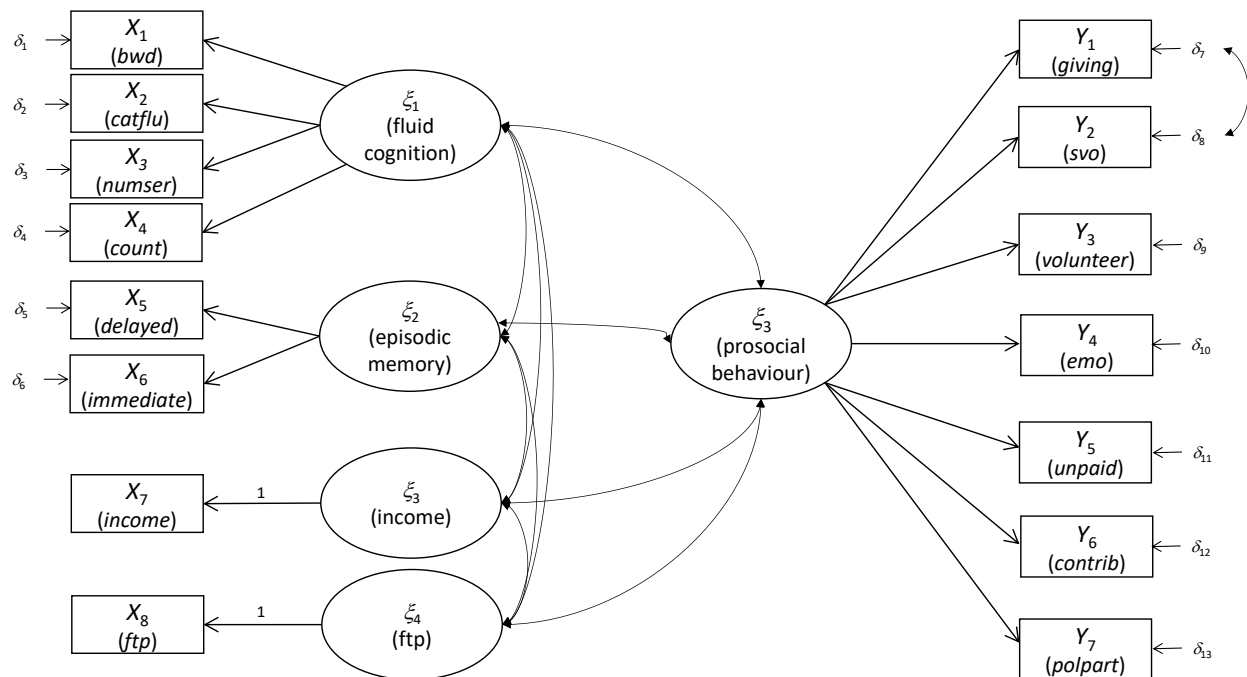


Figure S10.1 Prosocial behaviour structural regression model re-expressed as a CFA model

Two-step rule, step 2: Prosocial behaviour structural regression paths

The second step is to imagine the structural model portion of the full model as if the latent variables were observed variables without any measurement error. That is, treat the structural model among latent variables as though it were a path analysis model among observed variables. Figure S10.2 shows that the structural portion of the prosocial-behaviour model is identified because it is recursive.

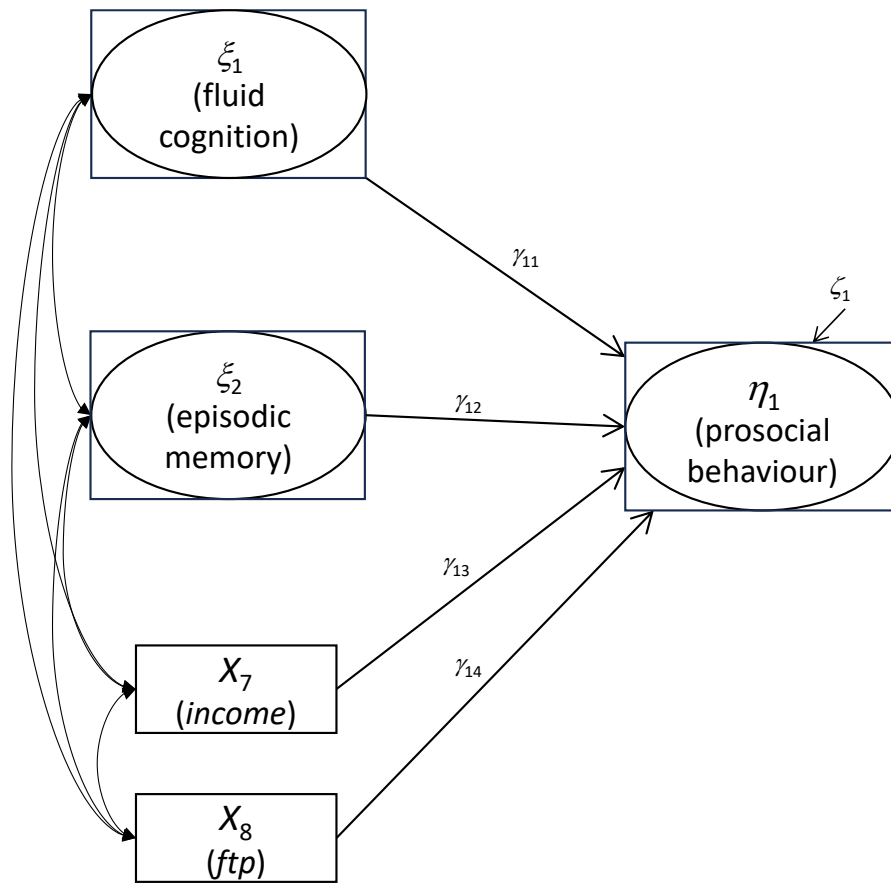


Figure S10.2 Prosocial behaviour structural regression model as a path analysis model